1. Given that vector $\boldsymbol{b}$ is collinear with $\boldsymbol{a} = 2\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k}$ and $\boldsymbol{a} \cdot \boldsymbol{b} = -18$, find the vector $\boldsymbol{b}$.
2. Given $\|\boldsymbol{a}\| = 4$, $\|\boldsymbol{b}\| = 2$, $\|\boldsymbol{a} - \boldsymbol{b}\| = 2\sqrt{7}$, find the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$.
3. Find the area of the triangle with vertices $A(1, -1, 1)$, $B(-1, 0, 2)$, $C(2, -2, 1)$, and the height on side AB.

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4. Given two points $M_1(3, -1, 2)$ and $M_2(4, -2, -1)$, find the plane through M_1 perpendicular to M_1M_2.
5. Find the angle between each of the following pairs of planes: $2x - 3y + 6z - 12 = 0$, $x + 2y + 2z - 7 = 0$
6. Find the distance between each of the following pairs of parallel planes: $19x - 4y + 8z + 21 = 0$, $19x - 4y + 8z + 42 = 0$
7. Compute the distance from the point to the plane: $M(1, 2, -3)$, $\Pi : 5x - 3y + z + 4 = 0$

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8. Find the equations of the following lines: (1) the line through the point $(1, 0, -2)$ perpendicular to both lines $\frac{x-1}{1} = \frac{y}{1} = \frac{z+1}{-1}$ and $\frac{x}{1} = \frac{y-1}{-1} = \frac{z}{0}$ (2) the line through the point $(2, -1, 3)$ that intersects the line $\frac{x+3}{7} = \frac{y}{0} = \frac{z+6}{2}$ perpendicularly
9. Find the equations of the following planes: (1) the plane through the point $(2, 0, -1)$ containing the line $\frac{x+1}{2} = \frac{y}{-1} = \frac{z-2}{3}$ (2) the plane containing the line $\frac{x-2}{1} = \frac{y+3}{-5} = \frac{z+1}{-1}$ and parallel to the line $\begin{cases} 2x - y + z - 3 = 0 \\ x + 2y - z - 5 = 0 \end{cases}$
10. Find the line through the point $(4, 0, -1)$ that intersects both lines $\begin{cases} x + y + z = 1 \\ 2x - y - z = 2 \end{cases}$ and $\begin{cases} x - y - z = 3 \\ 2x + 4y - z = 4 \end{cases}$.
11. Find the distance from the point $P(3, -1, 2)$ to the line $\begin{cases} x + y - z + 1 = 0 \\ 2x - y + z - 4 = 0 \end{cases}$.

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12. Write the equations of the surfaces of revolution obtained by rotating the following curves about the specified axes: (1) the circle $x^2 + z^2 = 9$ in the zOx plane rotated about the z axis (2) the parabola $z^2 = 2px$ in the zOx plane rotated about its axis of symmetry and about the tangent line through its vertex

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13. Find the equation of the cylinder through the curve $\begin{cases} x^2 + y^2 + 4z^2 = 1 \\ x^2 = y^2 + z^2 \end{cases}$ whose generators are parallel to the z axis.
14. Find the projections of the space curve $\begin{cases} x^2 + (y-1)^2 + (z-1)^2 = 1 \\ x^2 + y^2 + z^2 = 1 \end{cases}$ onto the three coordinate planes.